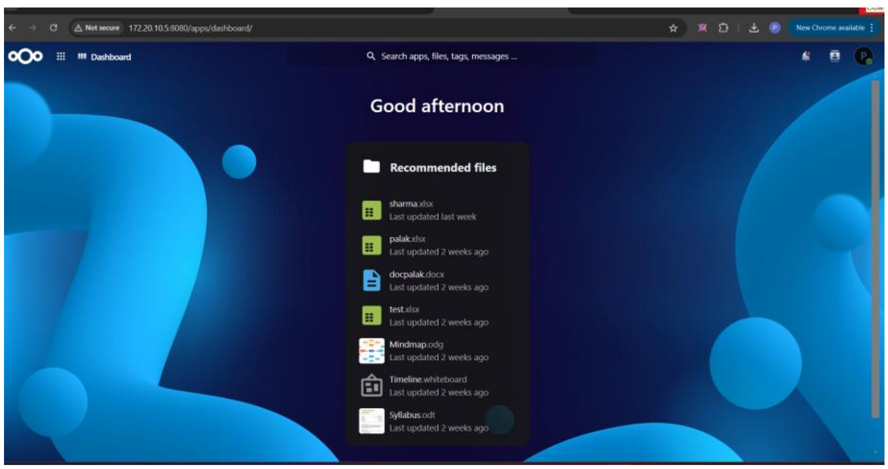
PROJECT REPORT

Palak_XLSV

Nextcloud-Based Offline Collaboration System for Real-Time Spreadsheet Editing



Nextcloud Dashboard running on the local server

Submitted By

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Open-Source File and Document Collaboration Platform

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NOTE: This report does not contain any confidential information about DRDO

1. Introduction

In most small organisations, colleges, and project teams, spreadsheet files are still shared through email attachments, USB drives, or third-party cloud storage services such as Google Drive or Dropbox. These approaches create version-conflict problems, depend on an active internet connection, and expose sensitive data to external servers that are outside the organisation's control. The Palak_XLSV project was undertaken to design and demonstrate a low-cost, fully self-hosted alternative that allows multiple users on the same local network to open, edit, and save the same Excel spreadsheet at the same time, without the data ever leaving the local environment.

The solution is built entirely on free and open-source software. Nextcloud, a widely used self-hosted file-sync-and-share platform, is combined with Collabora Online (an open-source office suite compatible with Microsoft Office formats) to provide live, multi-user spreadsheet editing inside a normal web browser. Both components are packaged as Docker containers and orchestrated through a single docker-compose.yml file, which makes the entire system reproducible on any machine that has Docker installed.

The finished system can run on a single laptop acting as a server. Once the containers are started, any device connected to the same Wi-Fi or LAN — laptops, desktops, or mobile phones — can open a browser, navigate to the host machine's local IP address, and collaborate on the same spreadsheet in real time, exactly like Google Sheets, but without using the public internet at any point.

1.1 Background

Offline or air-gapped collaboration is a recurring requirement in environments where internet access is restricted, unreliable, or undesirable for security reasons — examples include defence establishments, classified research labs, exam halls, and remote field offices. Existing cloud collaboration tools

(Google Workspace, Microsoft 365) cannot be used in such settings because they require continuous internet connectivity and route data through third-party servers. This project explores whether an equivalent collaborative experience can be reproduced purely on a local network using open-source software.

1.2 Why Docker and Nextcloud

Docker was selected because it removes the need to manually install, configure, and maintain a web server, PHP runtime, and database for Nextcloud. The entire application stack is described declaratively in a single configuration file and can be brought up or torn down with one command, which makes the project portable and easy to redeploy on any Windows, Linux, or macOS host. Nextcloud was selected over building a custom file-server because it already provides a mature web interface, user-account management, file versioning, and — most importantly — official integration with Collabora Online for real-time office-document editing.

2. Objectives

The project was carried out with the following specific objectives:

- To set up a self-hosted Nextcloud server using Docker that does not depend on any external/public cloud service.
- To integrate Collabora Online with Nextcloud so that Excel (.xlsx) spreadsheets can be opened and edited directly inside the browser.
- To enable simultaneous, real-time editing of the same spreadsheet by two or more users connected to the same local network.
- To make the entire setup reproducible through a single docker-compose.yml file, requiring minimal manual configuration.
- To verify that the system works entirely offline, i.e. without dependence on the public internet, once the local network and IP address are established.
- To document the complete workflow — from starting the containers to verifying multi-user collaboration — in a way that can be repeated by anyone with basic Docker knowledge.

2.1 Scope

The current scope of the project covers:

- Deployment of Nextcloud and Collabora Online as Docker containers on a single host machine.
- Discovery and use of the host machine's local-network IP address by client devices.
- Uploading, opening, and collaboratively editing spreadsheet (.xlsx) files through the Nextcloud web interface.

- Demonstration of two or more simultaneous editors working on the same sheet, with live cursor/selection indicators.

Future extensions such as HTTPS/SSL termination, LDAP-based authentication, persistent VirtualBox-hosted deployment, and mobile-app access are discussed in the Future Scope section of this report.

3. System Overview

Palak_XLSV is an offline collaboration system that combines container orchestration, a self-hosted file-sync server, and a browser-based office suite to deliver real-time spreadsheet editing on a local network. The system is composed of three logical layers, summarised below.

3.1 Conceptual Working

1. The host machine (a laptop, or alternatively a VirtualBox virtual machine) runs Docker Engine and Docker Compose.
2. Two containers are started from the compose file: one running the official nextcloud image, and one running collabora/code, the Collabora Online document server.
3. Once the host connects to the local Wi-Fi/LAN, it is assigned an IP address (e.g. 172.20.10.5) by the router/hotspot.
4. Any device on the same network can reach the Nextcloud web interface by browsing to http://<host-ip>:8080.
5. Users log in with their Nextcloud account, upload or open an existing .xlsx file, and the file opens inside Collabora Online's spreadsheet editor, embedded directly in the browser tab.
6. Because Collabora Online supports concurrent editing sessions, a second user opening the same file from a different device joins the same live editing session — both users see each other's cursor, selected cell, and edits in real time.
7. All changes are saved back to the spreadsheet file stored inside the Nextcloud Docker volume on the host machine; nothing is transmitted to the public internet.

3.2 High-Level Architecture

Layer	Component	Responsibility
Infrastructure	Docker Engine / Docker Compose	Builds, starts, and networks the application containers
Application	Nextcloud (nextcloud:latest)	File storage, user accounts, sharing, web dashboard

Office Engine	Collabora Online (collabora/code:latest)	Renders and edits Office documents (xlsx/docx/pptx) inside the browser, supports multi-user sessions
Network	Local Wi-Fi / LAN	Connects host and client devices without internet dependency
Storage	Docker named volume nextcloud_data	Persists uploaded files and the Nextcloud database/config across restarts

4. Technologies Used

4.1 Docker & Docker Compose

Docker is a containerisation platform that packages an application together with all its dependencies (libraries, runtime, configuration) into a single portable unit called an image. Docker Compose extends this by allowing several related containers — in this project, Nextcloud and Collabora — to be defined, networked, and started together using one declarative YAML file. This guarantees that the same environment can be reproduced identically on any machine, removing the 'it works on my computer' problem.

4.2 Nextcloud

Nextcloud is an open-source, self-hosted file-sync-and-share platform, comparable in functionality to Google Drive or Dropbox, but fully under the control of the organisation that deploys it. It provides a web-based dashboard, user and group management, file versioning, sharing links, and an extensible app ecosystem. In this project Nextcloud acts as the central storage and collaboration hub where spreadsheets are uploaded, organised, and opened for editing.

4.3 Collabora Online (Office Engine)

Collabora Online is an open-source online office suite built on LibreOffice technology. When connected to Nextcloud through the official integration, it allows Word, Excel, and PowerPoint format documents to be viewed and edited directly in a browser tab, with full support for concurrent multi-user editing, similar in spirit to Google Docs/Sheets. It is what gives Palak_XLSV its real-time, multi-user spreadsheet-editing capability.

4.4 VirtualBox (Optional Hosting Layer)

VirtualBox is a free, open-source hypervisor that creates an isolated virtual machine on top of a physical computer. In the deployment variant of this project, the Docker host (running Nextcloud and Collabora) is itself placed inside a VirtualBox VM rather than running directly on the physical laptop. When the VM boots, it obtains its own IP address from the local network (in bridged-adapter mode) exactly like a physical machine would. This isolates the collaboration server from the host operating system, makes the deployment portable as a VM image, and keeps the server running independently of what the host machine is otherwise used for. It is important to note that VirtualBox itself does not perform any collaboration — it is purely the hosting environment for the Docker containers.

4.5 YAML (docker-compose.yml)

YAML (Yet Another Markup Language) is a human-readable data-serialisation format used to describe the services, images, ports, volumes, environment variables, and restart policies for both containers in a single, version-controllable configuration file.

5. Docker Compose Configuration

The complete application stack is defined in a single docker-compose.yml file. This file declares two services — nextcloud and collabora — along with the named volume used for persistent storage. The configuration used in this project is reproduced below.

```
File    Edit    View                                   H1

version: ”3”
services:
  nextcloud:
    image: nextcloud:latest
    ports:
      - ”8080:80”
    volumes:
      - nextcloud_data:/var/www/html
    restart: unless-stopped

  collabora:
    image: collabora/code:latest
    ports:
      - ”9980:9980”
    environment:
      - domain=172\.20\.10\.5
      - username=admin
      - password=adminpass
      - extra_params=--o:ssl.enable=false --o:ssl.termination=false
    restart: unless-stopped
    cap_add:
      - MKNOD

volumes:
  nextcloud_data:
```

Figure 1 — docker-compose.yml defining the Nextcloud and Collabora services

5.1 Explanation of Key Configuration Fields

Field	Purpose
image: nextcloud:latest	Pulls the official Nextcloud Docker image from Docker Hub
ports: 8080:80	Maps container port 80 to host port 8080, so Nextcloud is reachable at http://<host-ip>:8080

volumes: nextcloud_data:/var/www/html	Persists Nextcloud's files, config, and database so data survives container restarts
restart: unless-stopped	Automatically restarts the container if it crashes or the host reboots
image: collabora/code:latest	Pulls the official Collabora Online (CODE) Docker image
ports: 9980:9980	Exposes the Collabora document-editing server on port 9980
environment: domain=172.20.10.5	Restricts Collabora to accept connections only from the trusted Nextcloud host (the local-network IP)
username / password	Sets administrator credentials for the Collabora admin console
extra_params: ssl.enable=false	Disables SSL inside the container since the system runs over plain HTTP on a trusted local network
cap_add: MKNOD	Grants the container the Linux capability required by Collabora to create device nodes during document conversion

Once this file is saved as docker-compose.yml, the Docker Desktop application must first be opened and left running in the background (this starts the underlying Docker engine). Only after Docker Desktop shows the engine as running can the entire stack be started from the Command Prompt with a single command:

```
docker compose up -d
```

Docker Compose then pulls both images (if not already present locally), creates an internal network so the two containers can communicate with each other by service name, creates the nextcloud_data volume, and starts both containers in the background.

6. Setup and Deployment Procedure

The deployment of Palak_XLSV follows a simple, repeatable sequence of steps, summarised in the table below and elaborated thereafter.

Step	Description
1	Install Docker Desktop (and optionally VirtualBox) on the host machine
2	Open/launch the Docker Desktop application and wait for the Docker engine to start (whale icon shows "running") — the containers cannot be started until Docker Desktop is open
3	Create the docker-compose.yml file with the Nextcloud and Collabora service definitions
4	Open Command Prompt and run "docker compose up -d" to build and start both containers
5	Connect the host machine to the local Wi-Fi/LAN network
6	Identify the host's local IP address (e.g. using ipconfig / ifconfig)
7	Open a browser on any client device and navigate to http://<host-ip>:8080
8	Complete the Nextcloud first-run admin setup (only required once)
9	Upload or create an Excel (.xlsx) spreadsheet inside Nextcloud
10	Open the spreadsheet — it loads inside the Collabora Online editor

11	Share the same local IP address with other users on the network so they can open and edit the same file simultaneously

It is important to note that Docker Desktop must be opened and fully running first — only after the Docker engine has started can any docker or docker compose command be executed successfully from the Command Prompt. All remaining steps, starting with creating the docker-compose.yml file and running "docker compose up -d", are then carried out through the Command Prompt (or any terminal) on the host machine.

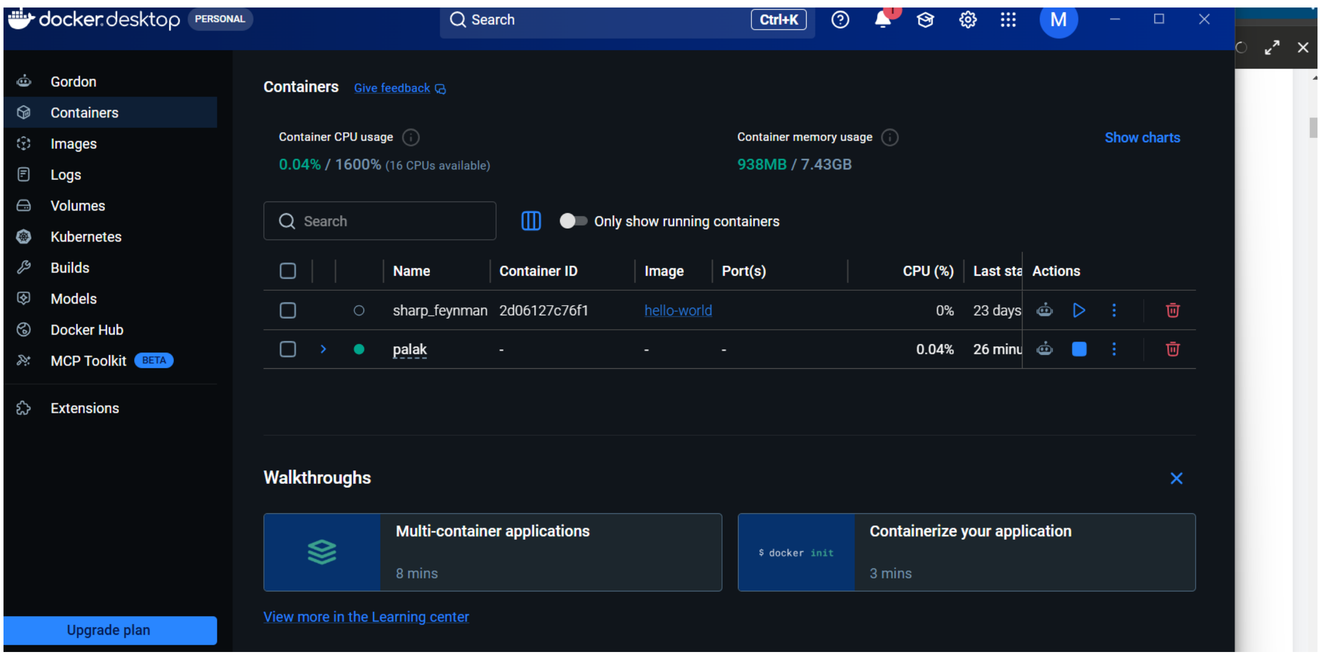


Figure 2 — Docker Desktop opened and running, with the "palak" container active under the Containers tab

6.1 Verifying the Running Containers

After running docker compose up -d, the status of both containers can be verified using the docker ps command, which lists the container ID, image, status, and exposed ports. In this project, the command confirmed that the palak-nextcloud-1 and palak-collabora-1 containers were both in the "Up" state, with port 8080 mapped for Nextcloud and port 9980 mapped for Collabora, as shown below.

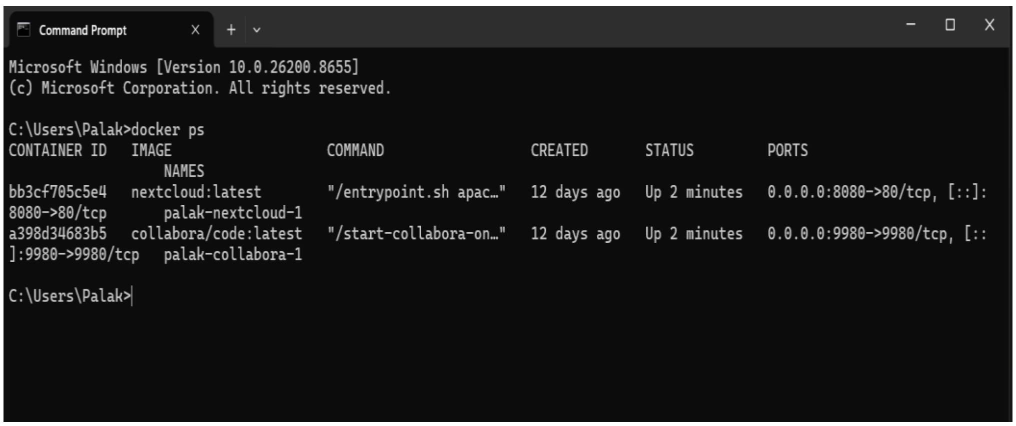


Figure 3 — Output of "docker ps" confirming both containers are running and healthy

6.2 Obtaining the Local IP Address

Because the system is designed to work without any external/public internet connection, the host machine's IP address on the local Wi-Fi network is what client devices use to reach Nextcloud. In this deployment, the host obtained the address 172.20.10.5 from the local hotspot/router, confirmed through the network settings of the host device, as illustrated below. This same IP address is then typed into the browser address bar (e.g. 172.20.10.5:8080) by every collaborating user.

- **dashboard.php** – Admin dashboard displaying intern applications and assignments.
- **assign_department.php** – Interface for admins to assign departments to interns.
- **ithardware_list.php** – Frontend page displaying IT hardware inventory.

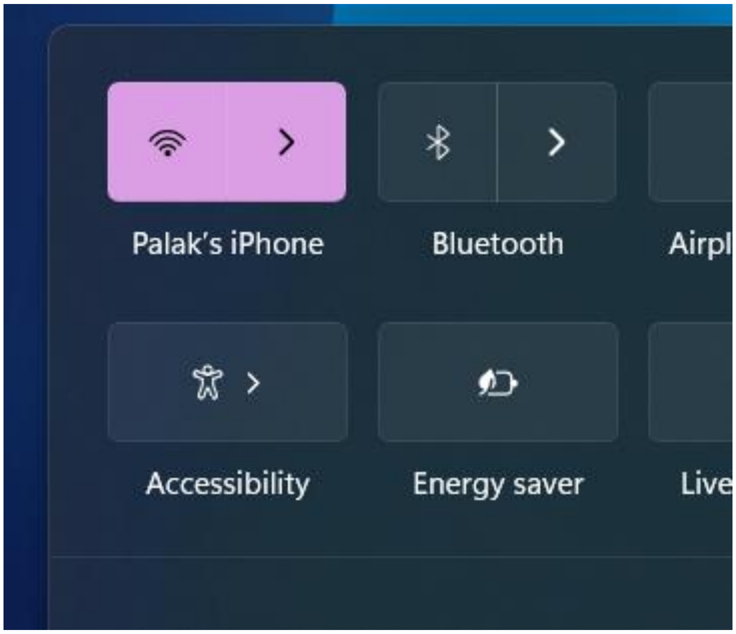
Scripts and Styles:

- **CSS Files** – Used across PHP files to style the layout, forms, tables, and buttons.
- **JavaScript** – Included for form validation or dynamic elements (not in uploaded files, but exist inline or separately).

Backend Logic:

- **connect.php** – Database connection logic.
- **register_action.php** – Processses intern registration form submission.
- **login_action.php** – Handles user authentication.
- **assign_department_action.php** –Login for assigning departments to interns.

- **Fetch_interns.php** – Fetches intern data for admin dashboard.
- **logout.php** – Handles user logout



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• *Figure 4 — Network panel confirming the active local Wi-Fi connection used to obtain the host IP address*

• 6.3 Nextcloud Login Screen

• Once Nextcloud is reachable at the host's local IP address, the very first screen presented to any user on the network is the login page (http://172.20.10.5:8080/login). This is the main entry point of the application — every collaborator, regardless of which device they connect from, must authenticate here with their Nextcloud account name and password before they can access the dashboard or any shared spreadsheet. The "Remember me" option allows a trusted device to stay signed in across sessions, and the "Log in with a device" option supports app-based or token-based authentication for client applications.

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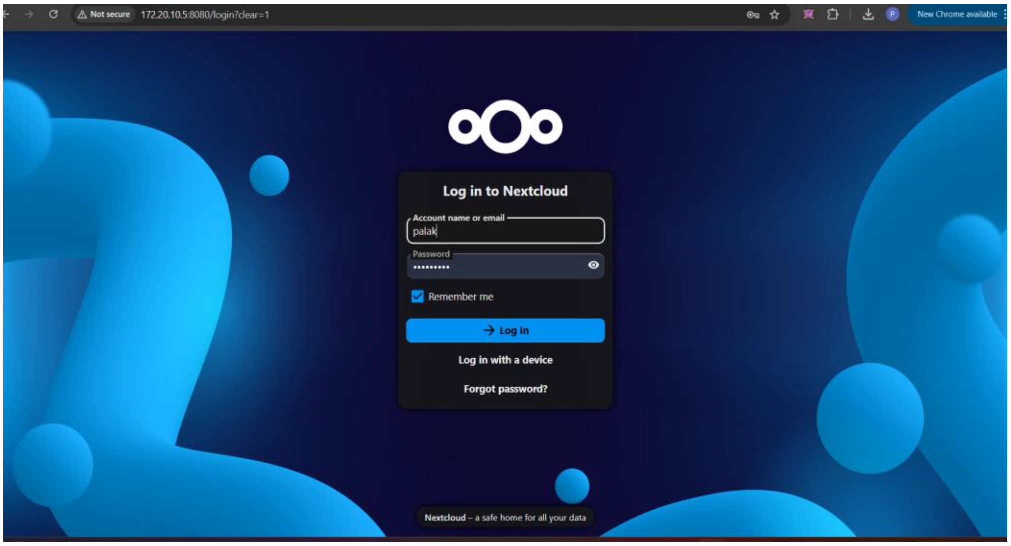

• *Figure 5 — Nextcloud login screen, the main entry point of the application, accessed at 172.20.10.5:8080*

• 6.4 Accessing the Nextcloud Dashboard

• After a successful login, browsing to the host's IP address on port 8080 (e.g. http://172.20.10.5:8080/apps/dashboard/) opens the Nextcloud web dashboard. The dashboard greets the logged-in user and displays a "Recommended files" panel listing recently modified documents, including the spreadsheets used for testing this project (sharma.xlsx, palak.xlsx, test.xlsx) along with other Nextcloud-native document types (docpalak.docx, Mindmap.odg, Timeline.whiteboard, Syllabus.odt). This confirms that the Nextcloud application layer is fully functional and reachable purely over the local network.

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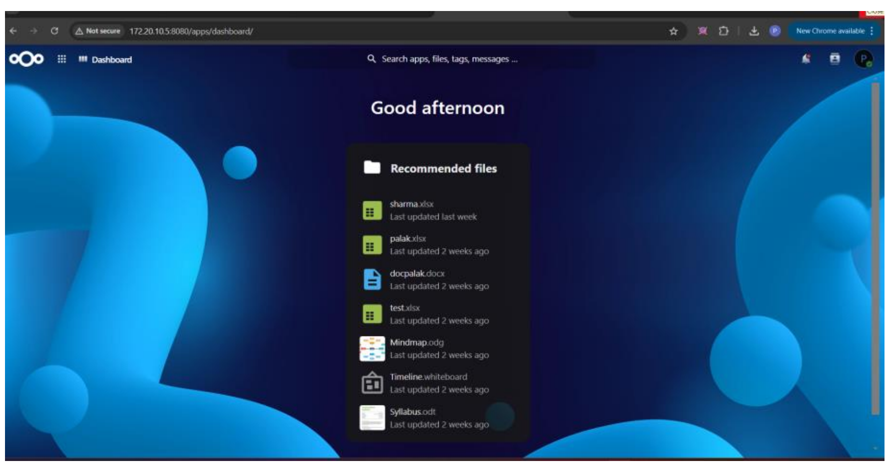

• *Figure 6 — Nextcloud dashboard accessed at the host's local IP address, showing recently used files*

7. Multi-User Real-Time Collaboration

The central goal of this project is to demonstrate that two or more users, connected to the same local network, can open and edit the same spreadsheet at the same time, with edits from one user instantly visible to the other. This was verified using the file sharma.xlsx, opened simultaneously from two different browser sessions while connected to the same Wi-Fi network as the host.

7.1 Opening the Shared Spreadsheet

The spreadsheet sharma.xlsx, containing a sample bar chart and a small data table (Category 1–3 against Object 1–5), was opened through the Nextcloud Files application. Selecting the file launches it directly inside the embedded Collabora Online spreadsheet editor in the same browser tab, with the full ribbon interface (Home, Insert, Page Layout, Formulas, Data, Review, View) available exactly as in a desktop spreadsheet application.

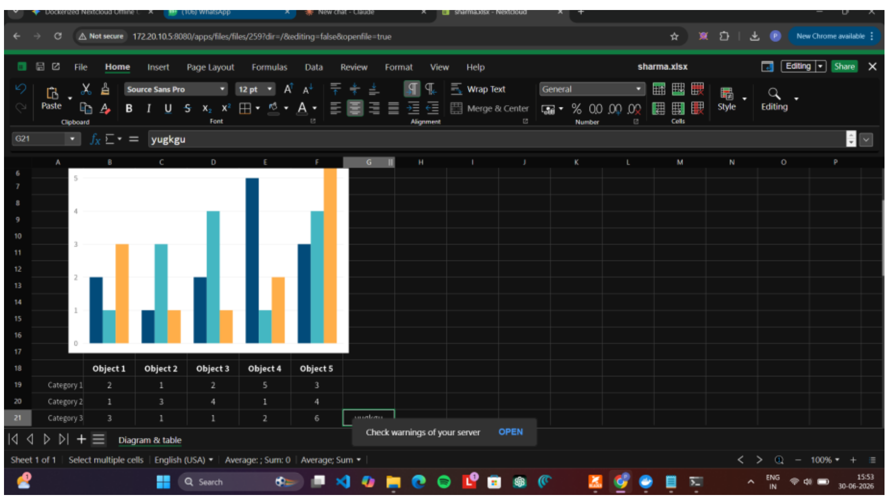

Figure 7 — sharma.xlsx opened for editing inside the Collabora Online spreadsheet editor

7.2 Simultaneous Editing by Two Users

When a second device, connected to the same local IP address, opens the identical file, Collabora Online automatically merges both sessions into one shared editing session. Each collaborator is shown with a uniquely coloured cursor/cell-selection box and a name label, so it is immediately clear who is editing which cell. In the test conducted for this project, two collaborators — labelled "disha" and "palak" in the editor — were simultaneously active inside the same sharma.xlsx file, each with their own highlighted cell selection, while the underlying chart and data remained synchronised for both.

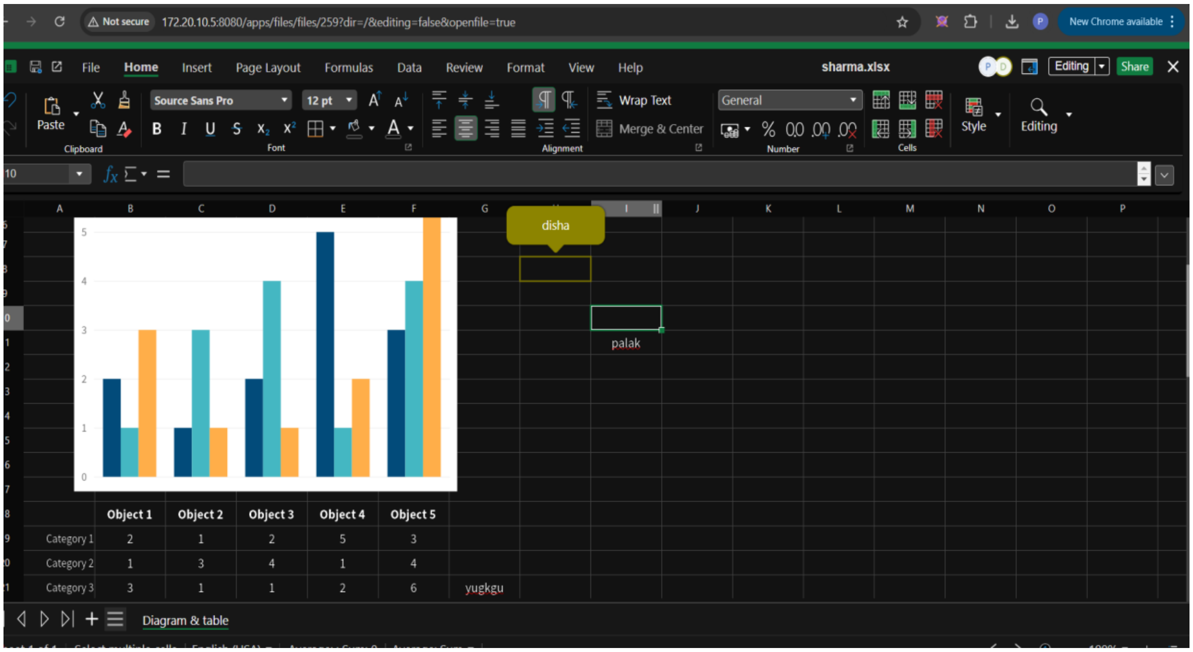

Figure 8 — Two users (disha and palak) editing the same spreadsheet concurrently, each with a distinct, colour-coded cell selection

7.3 Observations from the Test

- Both users were able to view, scroll, and edit the spreadsheet without conflicts or lock errors.
- Cell selections and live cursor positions of one user were visible in real time to the other user.
- Changes made by either user were saved to the same underlying .xlsx file stored on the Nextcloud server.
- The entire interaction worked using only the local IP address (172.20.10.5), confirming the offline / no-internet-dependency requirement of the project.
- No additional client software was required on either device — only a standard web browser.

8. Core Functionalities

- Containerised Deployment: Nextcloud and Collabora Online run as isolated Docker containers, started and stopped together via Docker Compose.
- Local-Network File Hosting: Spreadsheet and document files are stored entirely on the host machine inside a Docker volume — no external server is involved.
- Multi-User Real-Time Editing: Multiple users on the same network can open and edit the same Excel file concurrently, with live cursor and selection indicators.
- Browser-Based Access: No client installation is required; any modern browser on any device on the network can access the dashboard and editor.
- Persistent Storage: The nextcloud_data Docker volume ensures uploaded files and configuration survive container restarts.
- IP-Based Access Control: Collabora is configured to trust only the host's local IP address, restricting access to the local network.
- Recommended Files Dashboard: Nextcloud automatically surfaces recently modified files for quick access by all users.
- Optional VM Isolation: The entire Docker host can optionally run inside a VirtualBox virtual machine for additional isolation and portability.

9. Workflow

The end-to-end workflow of Palak_XLSV, from server start-up to collaborative editing, is summarised below.

1. Start the host machine and open the Docker Desktop application (and VirtualBox, if the VM-hosted deployment variant is used) — Docker Desktop must be fully running before any command-line step.
2. Open Command Prompt and run docker compose up -d to start the nextcloud and collabora containers.
3. Verify both containers are healthy using docker ps.
4. Connect the host to the local Wi-Fi/LAN network and note the assigned IP address.
5. Open the Nextcloud dashboard from a browser using http://<host-ip>:8080.
6. Log in with a Nextcloud account, or complete the first-time admin setup if running for the first time.
7. Upload an existing Excel file or create a new spreadsheet directly inside Nextcloud.
8. Open the spreadsheet — it loads inside the Collabora Online editor embedded in the browser.
9. Share the local IP address with other users on the same network.
10. Other users open the same file from their own devices and join the same live editing session.
11. All users edit the spreadsheet concurrently; changes are saved automatically to the shared file on the host.
12. Users log out / close the browser tab when finished; the host server keeps running, ready for the next session.

10. Challenges and Solutions

10.1 Connecting Collabora to Nextcloud

Challenge: Collabora Online, by default, only accepts editing requests from explicitly trusted domains/IP addresses, and any mismatch between the configured domain and the actual host IP results in a "loading document" failure inside the browser.

Solution: The Collabora container's environment variable was explicitly set to the host's local IP address (domain=172.20.10.5), and SSL was disabled (ssl.enable=false, ssl.termination=false) since the system operates entirely over plain HTTP within a trusted local network, avoiding certificate-related connection failures.

10.2 Network/IP Address Stability

Challenge: Local IP addresses assigned by a Wi-Fi router or mobile hotspot can change between sessions, which would break the trusted-domain configuration in Collabora and force users to use a new address each time.

Solution: For testing, a personal hotspot with a static/known IP behaviour was used; for a production deployment, a static IP reservation (DHCP reservation) on the router, or a fixed bridged-adapter IP inside the VirtualBox VM, is recommended so the address remains constant across reboots.

10.3 Container Capability for Document Conversion

Challenge: Collabora's document-conversion engine requires the ability to create device nodes inside its container, which is blocked by Docker's default security restrictions.

Solution: The cap_add: MKNOD capability was explicitly added to the Collabora service in the docker-compose.yml file, granting just the specific Linux capability required, without disabling container security altogether.

10.4 Persisting Data Across Restarts

Challenge: Without persistent storage, all uploaded files and Nextcloud configuration would be lost every time the container was recreated.

Solution: A named Docker volume, nextcloud_data, was mounted to /var/www/html inside the Nextcloud container, ensuring that all files, the database, and configuration persist independently of the container's lifecycle.

10.5 Verifying True Multi-User Concurrency

Challenge: It was necessary to confirm that two users editing the same file were genuinely sharing one live session (rather than each silently overwriting the other's changes on save).

Solution: The test was performed using two separate devices on the same network, opening the identical file at the same time; Collabora's live cursor and named-selection indicators (visible as "disha" and "palak" in the editor) confirmed that both users were inside the same real-time session.

11. Results and Observations

The Palak_XLSV system was successfully deployed and tested in a local-network environment. The following outcomes were observed during testing:

11.1 Successful Deployment

- Both the Nextcloud and Collabora containers started successfully and remained stable ("Up") over the testing period, confirmed via docker ps.
- The Nextcloud dashboard was reachable from multiple devices using only the host's local IP address, with no internet connectivity required.

11.2 Functional Spreadsheet Collaboration

- Excel (.xlsx) files could be uploaded, opened, and edited directly inside the browser using the Collabora Online editor.
- Two users were able to simultaneously edit the same spreadsheet, with each user's cell selection and changes visible to the other in real time.
- Formatting features (charts, fonts, number formatting, cell styling) inside the spreadsheet were preserved correctly during collaborative editing.

11.3 Security and Cost Benefits

- Since no data leaves the local network, the system is inherently more secure for sensitive or confidential spreadsheets compared to public cloud services.
- Because every component used (Docker, Nextcloud, Collabora Online) is free and open-source, the system has zero licensing cost.

11.4 Limitations Observed

- The system currently runs over plain HTTP; for production use beyond a trusted LAN, SSL/TLS should be enabled.

- IP-address changes on the host require the Collabora domain configuration to be updated, which is a manual step in the current setup.
- User-account management is currently done manually through the Nextcloud admin panel rather than integrating with an existing organisational directory (e.g. LDAP/Active Directory).

12. Advantages of the System

Aspect	Benefit
Cost	Built entirely on free, open-source software — no licensing fees
Security	Data never leaves the local network; no dependency on third-party cloud servers
Portability	The entire stack is defined in one docker-compose.yml and can be redeployed on any Docker-capable machine
Offline Operation	Functions fully on a local Wi-Fi/LAN network without internet access
Real-Time Collaboration	Multiple users can edit the same spreadsheet simultaneously, similar to Google Sheets
Familiar Interface	Collabora Online's ribbon UI closely resembles Microsoft Excel, requiring no retraining
Isolation (optional)	Hosting inside a VirtualBox VM keeps the server isolated from the host operating system

13. Future Scope

- Enable HTTPS/SSL termination using a reverse proxy (e.g. Nginx or Traefik) for encrypted traffic even within the local network.
- Configure a static IP / DHCP reservation for the host so the access address never changes between sessions.
- Integrate LDAP or Active Directory authentication for centralised, organisation-wide user management.
- Package the VirtualBox VM as a portable .ova image so the entire server can be imported and run on any machine in minutes.
- Add automated backup of the nextcloud_data volume to prevent accidental data loss.
- Extend support beyond spreadsheets to collaborative editing of Word and PowerPoint files, which Collabora Online already supports.
- Develop a simple companion mobile app or use the existing Nextcloud mobile client for easier access from phones and tablets on the local network.

14. Conclusion

The Palak_XLSV project successfully demonstrates that a fully functional, real-time, multi-user spreadsheet collaboration system can be built using entirely free and open-source software, without any dependency on the public internet or third-party cloud providers. By combining Docker Compose for reproducible deployment, Nextcloud as the file and collaboration hub, and Collabora Online as the in-browser office engine, the project achieves a Google-Sheets-like collaborative editing experience confined entirely to a local Wi-Fi/LAN network.

Testing confirmed that the system can be deployed with a single docker-compose up -d command, is reachable from any device on the same network using a simple local IP address, and supports genuine concurrent editing — verified by two simultaneous users ("disha" and "palak") editing the same sharma.xlsx file with live, colour-coded cursors. This makes the system a practical, low-cost, and secure alternative to public cloud collaboration tools, particularly suited to environments where internet access is restricted or where data confidentiality is a priority.

With the future enhancements outlined in this report — particularly SSL encryption, static IP allocation, and directory-based authentication — the system can be matured from a working prototype into a production-ready offline collaboration platform suitable for sustained organisational use.

- **MySQL Documentation –**
  URL: https://www.w3schools.com/sql/

This report presents a thorough examination of the training project, covering all aspects from the company profile to the system's implementation and outcomes. It offers an extensive analysis of the code files and their respective roles, ensuring a comprehensive understanding of the developed system.

Each section is meticulously crafted to clarify the processes involved, the technical decisions made, and the impact of the system on the organization's operations. This approach provides a holistic view of the project's achievements and its contributions to the organization's goals.